NIHAT CENGIZ PAMUK

Electrical & Electronics Engineer | Biomedical Engineer

Istanbul, Türkiye | +90 539 630 12 69 | nihatcengizpamuk@gmail.com

SUMMARY

Electrical & Electronics and Biomedical Engineer experienced in R&D, product development, prototyping, testing, and manufacturing. Contributed to a 40 kVA UPS system and a TÜBİTAK/KOSGEB-funded autonomous audiometry device. Skilled in PCB design (including Flyback SMPS, IGBT, and thyristor driver boards), test & verification, HMI touchscreen design, and certification readiness, alongside component procurement and order tracking.

SKILLS

- **Design Tools:** KiCad, Altium Designer
- **Power Electronics:** Flyback SMPS, IGBT gate driver circuits, UPS power stages
- **Embedded Systems & Communication:** STM32 microcontrollers, RS485, I2C, UART
- **Hardware Engineering :** PCB design, prototyping, EMI/EMC, electronic testing and validation
- **Languages:** Turkish (Native), English (Intermediate)

PROFESSIONAL EXPERIENCE

Electrical & Electronics Engineer **April 2023 – October 2026**

BetaUPS Kesintisiz Güç Kaynakları & Biensis Mühendislik

- Contributed to technical and operational processes from design to production in a 40 kVA UPS project.
- Carried out 2-layer PCB design and development for power electronic boards, including Flyback SMPS, IGBT gate drivers, and thyristor drivers.
- Executed electronic prototype assembly, hardware bring-up, testing, and validation processes.
- Designed touchscreen (HMI) interfaces and integrated RS485, I2C, and UART communication protocols.
- Managed Bill of Materials (BOM), component procurement, and order fulfillment workflows.

Co-Founder **September 2022 – November 2022**

AiHear Teknoloji A.Ş.

- Co-founded and contributed to the "Development of an Autonomous Audiometer Device" project funded by TÜBİTAK and KOSGEB.
- Played an active role in hardware development, testing, and regulatory certification processes for the medical device.

INTERNSHIPS

Intern **August 2021 – September 2021**

Canlar TM Tekstil Makina A.Ş.

- Contributed to the manufacturing, installation, and testing phases of industrial machinery for the textile sector.

Intern **August 2018 – September 2018**

3G Medikal

- Gained hands-on experience in technical support, installation, and marketing operations for endoscopy equipment.
- Conducted initial inspections and learned troubleshooting and fault-detection procedures for incoming service devices.

Intern **August 2017 – September 2017**

Istanbul Florence Nightingale Hospital (Şişli & Gayrettepe)

- Assisted in building and updating the medical device inventory management system.
- Participated in equipment troubleshooting, fault resolution, and scheduled preventive maintenance procedures.

EDUCATION

B.Sc. in Electrical and Electronics Engineering – *Zonguldak Bülent Ecevit University* **2019 – 2021**

B.Sc. in Biomedical Engineering – *Zonguldak Bülent Ecevit University* **2015 – 2019**

ADDITIONAL INFORMATION

Military Service: Completed | Driving License: Class B